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Maxwell equations, constitutive relations and electromagnetic boundary conditions

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Use macroscopic fields and free sources consistently:

$$\nabla \cdot \mathbf{D} = \rho_f, \qquad \nabla \cdot \mathbf{B} = 0,$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}.$$

For a fixed volume V with outward closed surface S, and a fixed open surface A bounded by an oriented contour C:

$$\oiint_S \mathbf{D} \cdot d\mathbf{a} = \int_V \rho_f \, dV, \quad \oiint_S \mathbf{B} \cdot d\mathbf{a} = 0,$$
$$\oint_C \mathbf{E} \cdot d\ell = -\frac{d}{dt} \int_A \mathbf{B} \cdot d\mathbf{a},$$
$$\oint_C \mathbf{H} \cdot d\ell = \int_A \mathbf{J}_f \cdot d\mathbf{a} + \frac{d}{dt} \int_A \mathbf{D} \cdot d\mathbf{a}.$$

The contour orientation follows the right-hand rule relative to the surface normal. Moving contours require the appropriate motional-emf treatment and are outside this session.

Derivation: apply the divergence theorem to each closed-surface law. Apply Stokes's theorem to each circulation law. Because each equality holds for arbitrary small regions in a smooth bulk medium, the integrands agree. At discontinuities, use the integral laws or a distributional interpretation instead of differentiating a discontinuous field as if it were smooth.

Physical interpretation: free charge sources D flux; B has no magnetic-charge source; changing magnetic flux induces E circulation; free current and changing D flux generate H circulation.

Units: E in V/m; D in C/m²; B in T; H in A/m; J_f in A/m²; ρ_f in C/m³. Do not mix microscopic total-source equations with macroscopic free-source equations.

displacement current and conservation

Take the divergence of Ampère–Maxwell law:

$$0 = \nabla \cdot \mathbf{J}_f + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) = \nabla \cdot \mathbf{J}_f + \frac{\partial \rho_f}{\partial t}.$$

Thus:

$$\boxed{\frac{\partial \rho_f}{\partial t} + \nabla \cdot \mathbf{J}_f = 0}.$$

The displacement term makes the curl equation compatible with changing charge density. It is not necessarily transport of free charges through an insulator.

Charging-capacitor example: neglect fringing in a parallel-plate vacuum capacitor of area A. D_n = Q/A, so the displacement current through the gap is I_d = A∂D_n/∂t = dQ/dt = I. Different spanning surfaces for the same Ampèrian loop give consistent circulation when the displacement term is included.

Numerical illustration: A = 0.010 m², gap d = 1.0 mm, dV/dt = 1.0×10⁶ V/s. C = ε₀A/d = 8.85×10⁻¹¹ F and I_d = C dV/dt = 8.85×10⁻⁵ A = 88.5 μA. Assumptions: uniform gap field, fixed plates, negligible fringing.

material response

Definitions:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}.$$

P describes electric dipole moment per volume; M describes magnetic dipole moment per volume. Bound volume charge is ρ_b = −∇·P. For a polarized body with outward normal n, bound surface charge is σ_b = P·n. Bound bulk current is J_b = ∂P/∂t + ∇×M, with the appropriate magnetization surface current at boundaries.

For a linear, local, isotropic, nondispersive material:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad \mathbf{M} = \chi_m \mathbf{H},$$

$$\boxed{\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad \mathbf{J}_{\text{cond}} = \sigma \mathbf{E}},$$

where ε = ε₀(1+χ_e) and μ = μ₀(1+χ_m). The free current may also contain impressed or other free-current contributions; do not identify all J_f with σE without a model assumption. Use σ only for conductivity and σ_f for free surface-charge density.

These are constitutive models, not additional universal Maxwell laws. In anisotropic material D_i = Σ_j ε_ij E_j. In dispersive material the response has time memory; in a frequency-domain model ε = ε(ω). Nonlinear response depends on field strength. Scalar component formulas used later require scalar isotropic ε and μ. Maxwell interface laws themselves are more general.

Local phasor extension, using e^(−iωt): ∇×E = iωB, ∇×H = J_f − iωD. For an Ohmic conductor with scalar σ and J_f = σE only, write ∇×H = −iωε_eff E with ε_eff = ε + iσ/ω. This incorporates conductivity once; do not add σE again if it is already absorbed in ε_eff. Treat this as a short extension, not the main lecture.

guided component calculation

At z = 0, medium 1 occupies z < 0 and medium 2 occupies z > 0. Define n = +$\hat{z}$. Let $\varepsilon_1 = 2\varepsilon_0$, $\varepsilon_2 = 5\varepsilon_0$, σ_f = 0 and $\mathbf{E}_1 = 30\hat{x} + 50\hat{z}$ V/m. Predict $\mathbf{E}_2$ before the boundary derivations.

After identifying tangential E continuity and normal D continuity, $\mathbf{E}_2 = 30\hat{x} + 20\hat{z}$ V/m. $D_1n = D_2n = 100\varepsilon_0$ C/m². E_n changes because ε changes, despite zero free surface charge. The total sheet charge inferred from the E jump is $\varepsilon_0(20-50) = -30\varepsilon_0$ C/m², entirely bound in this example.

derive the boundary conditions

Declare n from medium 1 to medium 2. Use stationary interfaces, ordinary finite bulk fields, and possible free electric surface charge σ_f and tangential free sheet current K_f. Exclude ideal dipole layers and singular magnetic-flux sheets from these standard derivations.

Normal D: Gaussian pillbox

For face area ΔA and height h, Gauss's law gives $(D_2n - D_1n)\Delta A$ plus side flux = enclosed free charge. Side flux vanishes as h→0 for finite fields; regular volume charge also vanishes, while σ_fΔA remains. Therefore:

$$\boxed{\hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f}.$$

Normal B: Gaussian pillbox

The same construction with $\oint \mathbf{B}\cdot d\mathbf{a} = 0$ gives:

$$\boxed{\hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0}.$$

Tangential E: shrinking rectangular loop

Use long sides of length $\Delta\ell$ tangent to the interface and short sides h normal to it. Faraday's law gives the difference of tangential line integrals. The magnetic-flux time derivative is proportional to $h\Delta\ell$ and vanishes as h→0 when $\partial\mathbf{B}/\partial t$ remains finite. Therefore:

$$\boxed{\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}}.$$

This condition is valid for ordinary time-dependent interfaces; it does not require electrostatics. Nonzero bulk curl E does not force a tangential interface jump.

Tangential H: shrinking rectangular loop

Ampère–Maxwell law retains a finite sheet-current contribution. Regular bulk current and displacement flux through the shrinking loop vanish. The vector result, with the declared normal, is:

$$\boxed{\hat{\mathbf{n}} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_f}.$$

Since K_f is tangential, $H_{2t} - H_{1t} = K_f \times n$. Check the sign by taking $n = \hat{z}$ and $K_f = K\hat{x}$: the H jump must be $-K\hat{y}$, because $\hat{z} \times (-K\hat{y}) = K\hat{x}$.

Component	General jump	Continuity condition
D normal	$D_{2n} - D_{1n} = \sigma_f$	$\sigma_f = 0$
B normal	$B_{2n} - B_{1n} = 0$	Ordinary interface, no magnetic charge
E tangential	$E_{2t} - E_{1t} = 0$	Ordinary finite-field interface
H tangential	$n \times (H_2 - H_1) = K_f$	$K_f = 0$

For scalar media: $\varepsilon_2 E_{2n} - \varepsilon_1 E_{1n} = \sigma_f$ and $\mu_2 H_{2n} = \mu_1 H_{1n}$. With $K_f = 0$, B_t can still jump if μ changes. With $\sigma_f = 0$, E_n can still jump if ε changes.

Graduate consistency extension: surface charge conservation at a planar stationary sheet is $\partial\sigma_f/\partial t + \nabla_s \cdot K_f + n \cdot (J_{f2} - J_{f1}) = 0$. Explain the signs: normal current leaving into medium 2 minus current entering from medium 1 contributes to charge depletion. For uniform steady conditions, $J_{f2n} = J_{f1n}$.

worked problems

Problem 1: charged dielectric interface

At $z = 0$ take $n = \hat{z}$, $\varepsilon_1 = 2\varepsilon_0$, $\varepsilon_2 = 4\varepsilon_0$, $E_1 = 100\hat{x} + 200\hat{z}$ V/m, and $\sigma_f = 3.542 \times 10^{-9}$ C/m². Find E_2.

Tangential continuity gives $E_{2x} = 100$ V/m. Normal D gives $4\varepsilon_0 E_{2z} - 2\varepsilon_0(200) = \sigma_f$. Since $\sigma_f/\varepsilon_0 \approx 400$ V/m, $E_{2z} \approx 200$ V/m. Thus $E_2 \approx 100\hat{x} + 200\hat{z}$ V/m. Equal normal E here is a consequence of the selected free sheet charge, not a universal boundary condition. $D_{2n} - D_{1n} \approx 400\varepsilon_0$ C/m² matches σ_f.

Problem 2: free current sheet

At $z = 0$ take $n = \hat{z}$, $K_f = 6\hat{x}$ A/m, $H_1 = 2\hat{y} + 3\hat{z}$ A/m, $\mu_1 = \mu_0$, $\mu_2 = 2\mu_0$. Find H_2.

Tangential jump: $H_{2t} - H_{1t} = K_f \times n = -6\hat{y}$ A/m, hence $H_{2y} = -4$ A/m and $H_{2x} = 0$. Normal B continuity: $2\mu_0 H_{2z} = \mu_0(3)$, hence $H_{2z} = 1.5$ A/m. Therefore $H_2 = -4\hat{y} + 1.5\hat{z}$ A/m. Verify $\hat{z} \times (H_2 - H_1) = 6\hat{x}$ A/m and $B_{2n} = B_{1n}$. These conditions determine the local interface mapping from given H_1; a global field still requires source and geometry data.

Paired extension if pace permits

Use the uncharged example $\varepsilon_1 = 2\varepsilon_0$, $\varepsilon_2 = 5\varepsilon_0$, $E_1 = 30\hat{x} + 50\hat{z}$. Let θ be the angle E makes with the normal. Derive $\tan\theta_2/\tan\theta_1 = \varepsilon_2/\varepsilon_1$ from E_t continuity and $\varepsilon_2 E_{2n} = \varepsilon_1 E_{1n}$. Values: $\theta_1 \approx 30.96°$, $\theta_2 \approx 56.31°$. This is a relation for electric-field direction at the interface, not optical Snell refraction.

perfect conductor and continuity with the previous lecture

For an ideal perfect electric conductor in the usual initially field-free model, E = 0 inside and B = 0 inside. Take medium 1 to be the conductor and n pointing outward into medium 2. Then:

$$\mathbf{E}_{2t} = 0, \quad D_{2n} = \sigma_f, \quad B_{2n} = 0, \quad \hat{\mathbf{n}} \times \mathbf{H}_2 = \mathbf{K}_f.$$

Do not claim that infinite conductivity alone expels a pre-existing static magnetic field. With E = 0, Faraday's law fixes ∂B/∂t = 0 inside, allowing trapped static flux. The usual B_n = 0 condition assumes zero internal B. Superconducting Meissner expulsion is a distinct physical statement.

In electrostatics E = −∇Φ. In a simple dielectric, Gauss's law becomes:

$$\nabla \cdot (\epsilon \nabla \Phi) = -\rho_f.$$

Only for uniform scalar ε does this reduce to ∇²Φ = −ρ_f/ε. At an ordinary interface without an ideal dipole layer, Φ₁ = Φ₂ and:

$$\epsilon_1 \frac{\partial \Phi_1}{\partial n} - \epsilon_2 \frac{\partial \Phi_2}{\partial n} = \sigma_f,$$

using the same n direction for both derivatives. These are the potential counterparts of tangential E and normal D conditions. Tangential E continuity fixes equality of tangential derivatives; ordinary potential continuity also requires exclusion of an ideal dipole sheet.

Preview only: in a homogeneous, source-free, lossless scalar medium, curl Faraday plus Ampère–Maxwell produces ∇²E − με∂²E/∂t² = 0, using ∇·E = 0. The interface conditions will determine reflected and transmitted amplitudes. Leave the full wave and Fresnel derivations to the next session.

Reading

- Jackson, Classical Electrodynamics, 3rd edition: Chapters 4 (dielectric response), 5 (magnetostatics and magnetic media), 6 (Maxwell equations), and 7 (plane waves and interfaces). This is a topic map, not a page or problem-number assertion.
- MIT OpenCourseWare, 6.641 Electromagnetic Fields, Forces, and Motion, Spring 2005, graduate lecture notes: L1–L3 for Maxwell equations and boundary conditions; L7–L8 for polarization, conduction and magnetization. https://ocw.mit.edu/courses/6-641-electromagnetic-fields-forces-and-motion-spring-2005/pages/lecture-notes/
- MIT OpenCourseWare, 6.013 Electromagnetics and Applications, Spring 2009, Lecture 2: Media and Boundary Conditions. https://ocw.mit.edu/courses/6-013-electromagnetics-and-applications-spring-2009/0ec2c39ffccb37e83b43f6f59868f8d7_MIT6_013S
- Richard Fitzpatrick, University of Texas at Austin, Classical Electromagnetism: Maxwell equations, dielectric media and magnetic media. https://farside.ph.utexas.edu/teaching/jk1/Electr

The external resources support derivations; their teaching order does not replace the official course outline. Modern SI note: μ₀ is approximately 4π×10⁻⁷ H/m, rather than an exact defining constant in the post-2019 SI; older notes may state otherwise.